

TABLES

Table 1. Characteristics of included data items ($n = 46$).

Author & Year	Design	Interv ention	Sample characteristics	Durat ion	Severity	Outcome measures	Main cognitive findings
(Aebi et al., 2020)	Non- randomised cross-over	NN, HH, NH, HN	16 healthy pilot trainees, 26 ± 4 yr	30 min	NN, HN = 440m HH, NH = 5500m	KLT KSS	$\uparrow$ KLT error in NH & HH $\downarrow$ Calculations in HH
(Ando et al., 2023)	Non- randomised cross-over	NN, HH	12 males, 23 ± 2 yr	7 hr	NN = 0.21 F _i O ₂ HH = 0.12 F _i O ₂	Go/No-Go Memory task accuracy	$\downarrow$ Memory task accuracy in HH $\leftrightarrow$ Go/No-Go RT & accuracy
(Asmaro et al., 2013)	Repeated measures within subject	NN, HH	35 aviation industry employees, 19-69 yr	30 min	NN = SL HH1 = 5335m HH2 = 7620m	ST DST Forward & Back TMT A & B	$\downarrow$ Performance during ST and DST Forward & Backward, & TMT A&B in HH

(Bailey et al., 2019)	Cross-sectional observational	-	23 male highlander 17 male lowlanders	-	Highlanders = 3640m Lowlanders = 80m	RAVL RDF & RDB TMT A & B DSST GPDT	↓ RAVLT, RDB, RDF, TMT B, DSST, ST in highlanders compared to lowlanders
(Bakharev, 1981)	Repeated measures within subject	LA, HA	7 young men	8 days	HA = 3400-3600m	Visual, short term, long term, operative, figurative, associative, involuntary, and auditory memory (RAVLT-like)	↓ Memory at day 2 Those of central Asia (highlanders) had less severe reductions compared to Pskov Region (lowlanders).
(Blogg & Gennser, 2006)	RCT	NN, HH	13 subjects, 29.5 ± 6.4	10 min	21kPa, 15kPa, 10kPa P_iO_2	FCRT Mannequin test Finger-tapping test Medium-tapping test	↔ Psychomotor performance at 15kPa ↑ FCRT and medium-tapping test error at 10kPa. Although

							participants perceived being more focused
(Bloomfield et al., 2023)	RCT	NN, PCH, ICH	12 adults, 26 ± 6.2 yr	30 min	10% oxygen gas (5500m)	CNSVS test battery	↓ Neurocognitive index score, cognitive flexibility, complex attention, composite memory, and visual memory in PCH ↓ Neurocognitive index score, composite memory, psychomotor speed and ↑ RT in ICH
(Bonnon et al., 1995)	Comparative	NN, HH	6 subject experimental and unspecified control group	60 hr	4383m	Cognitive-motor task	↑ Cognitive-motor performance in control group ↔ Performance in experimental group comparable once at 60 hrs.
(Bonnon et al., 1999)	Comparative	NN, HH	7 subject experimental and unspecified control group	17 days	9 days at 3500-4400m, 8 days climbing 500m/day to 5400m	Cognitive motor task	↔ Difference between experimental and control group.

(Bouquet et al., 1999)	Comparative	NN, HH	8 male climbers, 24 – 37 yr 10 controls, 22 – 40 yr	31 days	Simulated climb to 8848m	Purdue pegboard, recognition task	↓ Purdue pegboard performance > 8000m ↓ Recognition performance in from 5500m ↓ Performance in complex task vs simpler task
(Champod et al., 2013)	Non-randomised cross-over	NN, IH	17 healthy males	6 hrs of 1min on/1 min off	-	<i>n</i> -back task	↑ RT during spatial but not verbal n-back task in IH across all levels of difficulty
(Chroboczek et al., 2023)	Non-randomised cross-over	NN, NH	17 healthy males	15 min	NH = 0.13 F _i O ₂	ST	↔ ST reading and naming interference

(Crow & Kelman, 1971)	Repeated measures within subject	HH	86 medical students, 19 – 30 years. Organised into groups of 4 or 6	30 min	HH = 609m, 2438m, or 3656m	Required to memorise sequences of 6 digits at 1Hz (n-back-like task) to assess short-term memory	↔ error rate between conditions
(Davis et al., 2015)	Comparative	HH	132 residents into three groups based on altitude of residence (0-1500m, 1501-3000m, >3000m)	-	Testing occurred at 4860m	Go/no-go test score VFT Hand movement test	↔ VFT ↓ Go/no-go in >3000m group ↓ Correct hand movements in >3000m and 1500-3000m group compared to 1500m group
(Falla, van Veelen, et al., 2024)	RCT	NN, HH	48 pre-hospital emergency service personnel	30 min	NN = 200m HH = 3000m or 5000m	PVT BART DSST VAS	↓ PVT at 5000m compared to 200m, no difference in other test Participants did not detect ↓ PVT on VAS

(Falla, Hüfner, et al., 2024)	RCT	HH	36 pre-hospital emergency service personnel	30 min	Two trials at 4000m, one with supplement oxygen and one without (control)	PVT DSST <i>n</i> -back GPDT	↑ PVT with oxygen supplementation ↑ 0.37ms in PVT each % reduction in S _p O ₂ .
(Gerard et al., 2000)	Randomised cross-over	NH	23 healthy adults	2 hr	NH = 5000m with and without 6% O ₂ enrichment	Story Memory TMT A&B DSST Finger Tapping Digit Vigilance GPDT PASA Matrix Reasoning KSS Probed Memory Recall PVT	↓ RT in all tasks with O ₂ enrichment ↑ Hand-eye coordination, and mood with O ₂ enrichment.

						CPT	
						CTT	
						PERS	
						VAS	
(Guo et al., 2017)	Observation	NN, HH	163 healthy adults	24 hr	HH = 3700m	RT Attention DSST DST Pursuit aiming	↔ Cognitive measures between those with headache and those without at HA
(Hill et al., 2014)	Comparative	LA, HA	229 in total: children (4-10 yr, LA = 44, HA = 52), adolescents (13-16 yr, LA = 23, HA = 25), young adults (25-40yr, LA = 20, HA = 20), older adults (>50	-	LA = 500m HA = 3700m	Cognitive Drug Research battery	↓ Processing speed, attention in HA participants compared to LA.

years, LA = 22, HA =
23)

(Hohenauer et al., 2022)	Randomised cross-over	NN, NH, HH	20 healthy participants, 27.6 yr	NH = 15 min HH1 = 1hr HH2 = 24 hr	NN = 550m NH = .144 FiO ₂ HH = 2980m	ANAM	↔ RT, logical relations, math processing, switching, and <i>n</i> back tests in any condition
(Keeler et al., 2023)	Repeated measures within subject	NN, NH	21 healthy adults, 27 ± 5 yr	90 min	NN = 0.206 FiO ₂ NH = 0.14 FiO ₂	Gridshot Capacity Multitracker	↔ no difference in any gamified task.
(Koller et al., 1993)	Comparative	NH	10 acclimatised mountaineers and 11	-	NH = 6000m	Attentiveness	↓ Attentiveness in lowlanders compared to highlanders >4000m

		non-acclimatised					
		controls					
(Latshang et al., 2013)	Randomised cross-over	NN, HH	51 healthy men, 24 yr	2 days at each HH level	NN = 490m HH1 = 1630m HH2 = 2590m	PVT, DASS, <i>n</i> -back tasks	↔ All cognitive tasks across conditions.
(Legg et al., 2016)	Randomised cross-over	NN, HH	36 air force personnel	30 min	HH1 = 8000ft HH2 = 12000ft	ANAM	↔ All cognitive tasks across conditions
(Malle et al., 2013)	Comparative	NN, HH	28 pilots in HH and 29 control pilots	90 min	NN = SL HH = 10000m	PASA	↓ Working memory in HH group compared to NN
(Merz et al., 2013)	Comparative	NN, HH	32 mountaineers, separated into two groups summitting at different rates	-	NN = SL HH1 = 4497m HH2 = 5533m HH3 = 6265m	LB VFT RFFT CT	↔ All cognitive tasks across conditions

(Niedermeier et al., 2017)	RCT	NN, IH, HH	52 participants into a sham or IH pre-acclimatisation group	NN, IH = 1hr HH = 12 hr	NN = 0.2 F _i O ₂ IH = 0.13 F _i O ₂ HH = 0.13 F _i O ₂	Game of Dice Task	↓ Risky decision making in HH for IH group compared to sham
(Noble et al., 1993)	Comparative	NN, NH	2 groups of 12 male subjects	30 min	NN = SL NH = 78% S _p O ₂	TMT DSST LRT RT	↓ Performance of TMT and no learning effect in NH group compared to sham. ↑ RT
(Parker et al., 2017)	Comparative	NN, NH	10 orthopaedic surgeons into either NN or NH group	45 min	NN = SL NH = 0.14 F _i O ₂	VWMC	↔ VWMC tasks across conditions
(Pighin et al., 2014)	Randomised cross-over	NN, NH	26 university students, 23 ± 6.8 yr	60 min	NN = 0.21 F _i O ₂ NH = 0.14 F _i O ₂	Decision-making task	↓ Loss aversion in NH compared to NN.

(Quackenbush et al., 2016)	Comparative	NN, HH	1 group performed VIHT 1 group performed sham	30 min	HH1 = 3048m HH2 = 3658m	ST SDMT DST	↑ ST and SDMT performance for VIHT group compared to sham
(Ramírez-delaCruz et al., 2025)	Randomised cross-over	NN, NH	19 participants, 23.7 ± 3.9 yr	45 min	NN = SL NH1=0.15 F _i O ₂ NH2=0.13 F _i O ₂ NH3=0.11 F _i O ₂	Memory task, Go/No-go task (sustained attention), Ericksen Flanker test, BAT (anticipation), RT	↑ sustained attention at NH1, ↑ anticipation at NH1 & NH3, ↑ RT at NH3 ↔ no difference across conditions
(Rice et al., 2019)	Repeated measures within subject	NN, NH	60 aviation students	5 min/le vel	NN = SL NH1 = 2438m NH2 = 3658m NH3 = 4877m NH4 = 6096m	Visual Sequence Comparison X-Plane	↓ Visual Sequency Comparison & X-Plane performance in NH compared to NN

(Schega et al., 2016)	RCT	NN, IH	34 participants, 60 – 75 yr	90 min	IH = 90-85% S _p O ₂	ST	↑ ST performance in IH group
(Schega et al., 2013)	RCT	NN, IH	34 participants, 60 – 75 yr	90 min	IH = 90-85% S _p O ₂	ZVT	↑ ZVT performance in IH group
(Schlaepfer et al., 1992)	Repeated measures within subject	NN, HH, NH	10 healthy subjects	-	HH = 3450m NH = 0.145 F _i O ₂	Read briefly displayed letters	↓ time needed to read letters in HH and NH
(Shukitt-Hale et al., 1998)	Repeated measures within subject	NN, HH	23 army personnel, 18-29 yr	4.5 hr	NN = 550m HH1 = 4200m HH2 = 4700m	Battery of cognitive tasks	↓ Performance in battery of cognitive tasks only after a few hours of exposure in a grade dependent fashion
(Tesler et al., 2015)	Randomised cross-over	NN, HH	26 participants, 26.9 yr	2 days	NN = 490m HH1 = 1630m HH2 = 2590m	Visio-motor learning task	↓ Sleep dependent memory performance in HH conditions

(Thomas et al., 2007)	Prospective	HH	11 subjects, 27 ± 1.5 yr	9 hr	HH1 = 7700ft HH2 = 10000ft HH3 = 13000ft	PVT <i>n</i> -back	↔ No difference in PVT & <i>n</i> -back
(Turner, Byblow, et al., 2015)	Randomised match-paired	NN, NH	22 adults, 23 yr	50 min	NN = SL NH = 0.1 F _i O ₂	CNS Vital Signs	↓ Performance in CNS Vital Signs
(Turner, Barker-Collo, et al., 2015)	Randomised match-paired	NN, NH	15 adults, 21-55 yr	90 min	NN = SL NH = 0.1 F _i O ₂	CNS Vital Signs	↓ Performance in CNS Vital Signs
(van der Post et al., 2002)	Randomised cross-over	NN, NH	12 healthy subjects	2 hr	NN = 97% S _p O ₂ HH1 = 90% S _p O ₂ HH2 = 80% S _p O ₂	Memory task RT	↑ RT in HH2 compared to HH1 and NN. ↑ Cognitive performance in HH1 compared to NN and HH2

(Van Dorp et al., 2007)	RCT	NH	22 healthy subjects	45 min	Sham hypoxia = 103 mmHg PetO ₂ without inspired CO ₂ Isocapnic hypoxia = 40mmHg with 0.4 mmHg clamped CO ₂ Poikilocapnic hypoxia = 40mmHg without CO ₂	PVT and MAT-bat	↓ Performance in PVT and MAT-bat
-------------------------	-----	----	---------------------	--------	--	-----------------	----------------------------------

(Wu et al., 2023)	RCT	NN	82 males 20.76 ± 1.6,	6 days	HH = 3680m	WHO-NCTB	↑ Spatial memory and digit symbol performance in RIPC group compared to controls
		HH	into control or a RIPC group				

(Yan et al., 2011a)	Comparative	NN	2 groups: 1 born at SL, the other in HA	-	-	2-back task	↔ Working memory performance
(Yan et al., 2011b)	Comparative	NN	2 groups: 1 born at SL, the other in HA	-	-	Verbal working memory	↑RT in HA group ↓ Accuracy in HA group

Green arrows represent positive outcomes, yellow arrows represent no change in outcomes, red arrows represent negative outcomes. NN, normobaric normoxia; HH, hypobaric hypoxia; NH, normobaric hypoxia; HN, hypobaric normoxia; KLT, Konzentrations Leistungs Test-Revidierte Fassung; KSS, Karonlinska Sleepiness Scale; F_iO₂, fraction of inspired oxygen; RT, reaction time; RAVL, Rey Auditory Verbal Learning; RDF, Repetition of Digits Forward; RDB, Repetition of Digits Backwards test; TMT, Trail Making Test, DSST, Digit Symbol Substitution Test; ST, Stroop test; GPDT, grooved pegboard dexterity test; MoCA, Montreal Cognitive Assessment; BDI, Becks Depression Inventory; DST, Digit Span test; LA, low altitude; HA, high altitude; RCT, randomised control trial; P_iO₂, pressure of inspired oxygen; PCH, poikilocapnic hypoxia; ICH, isocapnic hypoxia; CNSVS, Central Nervous System Vital Signs; IH, intermittent hypoxia; VFT, verbal fluency test; BART, balloon analogue risk task; VAS, visual analogue scale; PASA, Paced Auditory Serial Addition; PVT, Psychomotor Vigilance Task; CPT, Calculation Performance Task; Compensatory Tracking Task; Performance Effort and Rating Scales; ANAM, Automated Neuropsychological Assessment Metrics; DASS, divided attention steering simulator; LB, line bisection test; RFFT, Ruff figural fluency test; CT, modified Pegboard test the Chess test; RBANS, The Repeatable Battery for the Assessment of Neuropsychological Status; WAIS-III, Wechsler Adult Intelligence Scale; RVP, Rapid Visual Information Processing test; OTS, One Touch Stockings of Cambridge; AMS, Acute Mountain Sickness; LRT, Letter Rotation Test; VWMC, verbal working memory capacity; VIHT, Voluntary Isocapnic Hyperventilation Training; SDMT, Symbol Digit Modalities Test; BAT, Bassin Anticipation Timer; MMSE, Mini Mental State Examination; ZVT, Number Combination Test; MAT-bat, Multi-Attribute Task battery; RIPC, remote ischemic precondition; WHO-NCTB, World Health Organization Neurobehavioral Core Test Battery; SL, sea level; FCRT, four choice reaction time.

Table 2. ROBINS-I-V2 risk of bias assessment for data items.

Data item	Domain 1	Domain 2	Domain 3	Domain 4	Domain 5	Domain 6	Domain 7	Overall
<i>Chroboczek et al., 2023</i>	Serious	Low	Low	Low	Moderate	Low	Moderate	Serious
<i>Crow et al., 1971</i>	Moderate	Moderate	Moderate	Low	Low	Low	Moderate	Moderate
<i>Gerard et al., 2000</i>	Serious	Low	Low	Low	Serious	Low	Moderate	Serious
<i>Guo et al., 2017</i>	Serious	Low	Moderate	Low	Low	Serious	Moderate	Serious
<i>Quackenbush et al., 2016</i>	Moderate	Serious	Moderate	Serious	Low	Low	Moderate	Serious
<i>Van Dorp et al., 2007</i>	Serious	Low	Low	Low	Low	Low	Moderate	Serious

Domain 1, Bias due to confounding; Domain 2, Bias in the selection of participants; Domain 3, Bias in the classification of interventions; Domain 4, Bias due to deviations from intended interventions; Domain 5, Bias due to missing data; Domain 6, Bias in the measurement of outcomes; Domain 7, Bias in the selection of the reported result.

Table 3. List of confounding factors taken into consideration of data items assessed against Domain 1 of the ROBINS-I V2.

Participant factors	Task factors	Environmental factors	Methodological factors
- Differences in baseline cognitive performance between conditions	- Complexity of task	- Severity and duration of hypoxia	- Sensitivity and reliability of tools used
- Age variability	- Familiarity to task	- Simulation or environmental hypoxia	- Timing of assessments
- Physical fitness variability		- Temperature and pressure	- Control groups used
- Health conditions			
- Acclimatisation			
- Genetic factors			

Figure 1

Forest plot of the overall effect of hypoxia on cognitive ability



